where $\Omega_i = k^{-1/2}\mu_i^{-1/2}$ is the frequency of the ith oscillator. Integrating by parts, we have

$$\int_0^t \Omega_i \sin(\Omega_i(t-s))Q(s)\mathrm{d}s = [\cos(\Omega_i(t-s))Q(s)]_0^t - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s$$

$$= Q(t) - \cos(\Omega_i t)Q(0) - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

which leads to

$$q_i(t) - Q(t) = \cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i/\mu_i - \cos(\Omega_i t)Q(0) - \int_0^t \cos(\Omega_i(t-s))P(s)\mathrm{d}s.$$

Introducing this formula in (6.23), we obtain a reduced system

$$\dot{Q} = P, \tag{6.27}$$

$$\dot{P} = -U'(Q) - k^{-1}\sum_{i=1}^{k}(Q - q_i), \tag{6.28}$$

$$= -U'(Q) + f(t) - \int_0^t \psi(t-s)P(s)\mathrm{d}s, \tag{6.29}$$

where we define

$$f(t) = -k^{-1}\sum_{i=1}^{k}\left[\cos(\Omega_i t)q_i(0) + \Omega_i^{-1}\sin(\Omega_i t)p_i(0)/\mu_i - \cos(\Omega_i t)Q(0)\right],$$

$$\psi(t) = k^{-1}\sum_{i=1}^{k}\cos(\Omega_i(t)).$$

For large k and for many choices of the frequencies Ω_i and the initial conditions $\{q_i(0), p_i(0)\}$, function $f(t)$ will take on the character of a Wiener process; in fact the convergence can be shown in the weak sense. We may even take the initial conditions to be randomly distributed [13, 205]. The function $\psi(t)$ will typically exhibit an initial decay followed by rapid bounded fluctuation (see Fig. 6.4), with decay to zero in the limit of large k. Let $\{\Omega_i^{(k)} | i = 1, 2, \ldots, k\}$ represent the frequencies used for a given k. We may make various proposals for the $\Omega_i^{(k)}$,